SeungEun Chung | 정승은

AI Engineer

(+82) 010-8590-3277 | seungeun.chung0222@gmail.com | linkedin.com/in/seung-eun-chung

Professional Summary

- Adaptable AI researcher specializing in reasoning segmentation with MLLM, with experience in audio-vision multimodal learning.
- Passionate about solving AI challenges, with hands-on experience in data acquisition, preprocessing for real-world applications.
- Quick to master new hardware and frameworks, and effective at solving complex challenges with a data-driven approach.

Education

Master's degree of Electronic Engineering 2024.03 – 2026.02
Kookmin University, Korea
- **Research**: Specialized in reasoning segmentation via LLMs and multimodal AI for real-world applications.
- **Selected Coursework**: AI Methodologies, Math & Optimization for AI, Advanced Intelligent Systems, Automotive Embedded SW, Advanced Automotive Intelligence, DSP Applications, AR/VR/XR.

Bachelor of Electronic Engineering 2020.03 – 2024.02
Kookmin University, Korea
- Technical Achievement Scholarship (2 semesters).

Publication

S. Chung, et al., "Research on Enhancing Reasoning Segmentation Performance via MLLM-based In-Context Learning," *The 38th Workshop on Image Processing and Image Understanding (IPIU)*, 2026.

S. Ahn, **S. Chung**, et al., "GuiFiR: Weakly and Semi-Supervised Learning of Hand Pose Estimation for Guitar Finger Reading," *IEEE Multimedia*, 2025. (Under Review)

S. Chung, et al., "Object Detection Method Using Frame Partitioning for Vehicle License Plate Detection," *The 37th Workshop on Image Processing and Image Understanding (IPIU)*, 2025.

Research Experience

Reasoning Segmentation through Prompting Large Language Models 2025.05 – Current
- **Requirement Development**: Developed a pipeline to translate complex, abstract textual instructions into simplified machine-executable queries using CoT (Chain-of-Thought) prompting and In-Context Learning.
- **Technical Depth**: Investigated spatially aware semantic segmentation behavior in MLLMs, demonstrating the ability to manage and explain high-level AI reasoning pathways.

Audio-vision Multimodal Learning for Guitar Finger Reading (GuiFiR) 2023.09 – 2025.04
- **Technical Problem Solving**: Solved a critical domain-specific data scarcity problem by building an automated labeling pipeline using Grounding DINO, processing 5,800 images in just 3 hours.
- **Outcome**: Proposed a novel multimodal task and achieved SOTA performance (MAE-H 9.372), with the work currently under review for IEEE Multimedia.

Frame Partitioning for License Plate Detection on Edge Devices 2024.09 – 2025.02

- **Requirement Management**: Analyzed industry partner's needs for high-resolution (4K) real-time processing on resource-constrained edge hardware.
- **Technical Bridge**: Designed a frame partitioning algorithm to bridge the gap between model accuracy and hardware limitations, improving F1-score from 88.25% to 93.48%.
- **Delivery**: Independently built a DeepStream-based pipeline and delivered the end-to-end system on schedule to meet strict industry partner requirements.

Development of an AI-based imaging skin diagnosis module using skin microbiome data 2023.09 – 2023.12

- **Stakeholder Alignment**: Conducted regular review meetings with the partnering company to align technical model outputs with the business goal of microbiome-based skin scoring.
- **Explainable AI (XAI)**: Implemented Grad-CAM visualization to provide clear, explainable diagnostic evidence, effectively communicating AI logic to non-technical stakeholders and ensuring client trust.
- **Outcome**: Achieved an average accuracy of 82.57%, surpassing the 80% target set by the partner, and successfully delivered the complete module.

Development of a facial expression information transmission system for the visually impaired 2023.03 – 2023.07

- **Global Communication**: Awarded Grand Award (1st Place) at ASCC 2023 (Bangkok) for innovation and "clarity in global research communication" in an English-speaking environment.
- **User-Centric Design**: Designed and evaluated a wearable system based on usability and comfort, proving the ability to interpret technical functions into user-centered benefits.

Award

Excellent Poster Presentation Award - IPIU 2026 (The 38th Workshop on Image Processing and Image Understanding) 2026.02

Grand Award (1st Place) - Asian Students' Capstone Design Contest (ASCC 2023) 2023.09
Bangkok, Thailand

1st Place Award - ASCC 2023 Korean Preliminary Competition 2023.08
Busan, Korea

Excellence Award (2nd Place) - Capstone Design Contest, Kookmin University 2023.08
Seoul, Korea

Bronze Prize - Electronic Engineering Design Competetion, Kookmin Universtiy 2023.12
Seoul, Korea

Bronze Prize - Electronic Engineering Design Competetion, Kookmin Universtiy 2023.01
Seoul, Korea

5th Place Award - Korean Institute of Electrical Engineers conference, Smart Energy Contest 2022.07
Yeosu, Korea

Language

Korean	***Native***	English	***Fluent*** (OPIC – Intermediate High)